

1. Briefly describe the differences between UDP and TCP.
2. What is a subnet mask? What is the subnet mask of 192.16.3.65/23?
3. Consider distributing a file of $F = 30$ Gbits to N peers. The server has an upload rate of $u_s = 30$ Mbps, and each peer has a download rate of $d_i = 2$ Mbps and an upload rate of $u = 700$ Kbps. Give the minimum distribution time for $N = 10$ and $N = 1,000$ for both
 - (1) client-server distribution;
 - (2) P2P distribution.
4. Suppose that the Go-back-N protocol is used. The sequence number space is $\{0, 1, 2, 3, 4\}$. Suppose the sender window size is 4. The sender sends packets 0, 1, 2 and 3. The receiver receives all 4 packets in order and generates acks. Suppose that all acks are lost.
 - (1) After the receiver sends out all acks, what will be the expected sequence number at the receiver side?
 - (2) After the timeout interval for packet 0, the sender retransmits packet 0. Suppose that the receiver receives the packet. What will the receiver do?
5. Consider sending a large file from a host to another over a TCP connection that has **no loss**. Suppose TCP's congestion control **does not use** slow start. We assume that **cwnd** increases by 1 MSS every time a batch of ACKs is received and that the round-trip time is constant.
 - (1) How long does it take for **cwnd** to increase from 10 MSS to 22 MSS assuming no loss events?
 - (2) What is the average throughput (in terms of MSS and RTT) for this connection during the time period that **cwnd** increases from 10 MSS to 22 MSS?
6. Suppose that an enterprise obtains a block of IP addresses where the network address is 201.16.18.0/23. The enterprise would like to partition the network into 5 subnets where the number of hosts to be supported in these 5 subnets are as follows: 233, 120, 50, 30, 14. Show your design of address allocation.